ISMAIL AMMAR

AI Engineer

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I'm an AI engineer. I build AI microservices and RAG / vector search pipelines in Python, on a background in systems-level C and C++ on Linux and in offensive security: I sit on the board of M4SEC, the CTF team ranked #1 in Morocco for the 2025 season, and compete in web security and DFIR.

SKILLS

AI Engineering	Python, FastAPI, RAG / vector search, pgvector, Ollama (local LLMs), PyTorch, scikit-learn, Pydantic, SQLAlchemy, PostgreSQL
Systems & DevOps	C, C++98, Linux, POSIX sockets, poll(), pthreads, Docker, Docker Compose, Nginx, Make, Git
Offensive Security	Web security, bug bounty / CTF, DFIR, network protocols (TCP, IRC)
Web	TypeScript, React, Node.js, Express, MongoDB, Tailwind CSS

PROJECTS

ft_transcendence | product owner and AI service, 1337 team of five, in progress — `github.com/yel-alja/FT_TRANSCENDENCE`

I build the AI service of ShieldCore, a bug bounty platform: duplicate detection, pre-triage and CVSS severity.

`Python · FastAPI · PostgreSQL · pgvector · Ollama · Docker`

BioGuide | NASA Space Apps Challenge 2025, team project — `github.com/Is-Ammar/BioGuide`

With my team, I built a web app for exploring NASA space biology research: search, dashboards and an AI chat assistant.

`React · TypeScript · Node.js · Express · MongoDB`

NavGuard | personal project — `github.com/Is-Ammar/navguard`

I built an anomaly detector for ship networks: it watches GPS, AIS and engine traffic for spoofing and tampering.

`Python · scikit-learn · PyTorch · Docker`

ft_irc | 1337 team project — `github.com/Is-Ammar/ft_irc`

With two teammates, I built an IRC server that serves many clients from one thread with non-blocking sockets and poll().

`C++98 · Sockets · poll() · TCP`

Minishell | C — `github.com/Is-Ammar/MINI-SHELL`

I wrote a Bash-like shell with pipes, redirections, heredocs and subshells.

Inception | Docker — `github.com/Is-Ammar/inception`

I set up WordPress, Nginx and MariaDB in Docker Compose, with every image built from scratch.

Other: I also built miniRT (ray tracer in C), Philosophers (threads and mutexes), push_swap, Minitalk and Cloud Wishes (CTF web challenge).

EXPERIENCE

Software Engineering Program | 1337 School (42 Network), Rabat — 2024 - Present

I'm in a project-based, peer-reviewed program: C, C++, Unix system programming, networking and Docker.

Board Member, Recruiter, Web & DFIR Player | M4SEC CTF Team — 2024 - Present

We ranked #1 in Morocco for the 2025 season and #135 worldwide on CTFtime, with top 10 finishes at CyberSphere (1st), L3akCTF (4th), DeadSec (5th), RSTCON (4th) and FortID (6th).

Freelance Developer | Web & Security — 2025 - Present

I build web apps and APIs for clients (React, TypeScript, Node.js, PostgreSQL) and review the security of their apps.

Navigation Systems Lead | Royal Moroccan Navy — 2022 - 2024

I ran shipboard navigation systems and telemetry, kept the sensor network at 99.9% uptime at sea, and led small teams.

Technical Analyst (Intern) | Royal Moroccan Navy — 2020 - 2022

I maintained and calibrated integrated bridge systems, and kept documentation in line with maritime regulations.

EDUCATION

1337 School (42 Network), Software Engineering — 2024 - Present
Hassan First University, Economics and Management — 2019 - 2021
Royal Moroccan Navy Training Center, maritime certifications — 2020 - 2023

LANGUAGES

Arabic, English, French